Quick Start Guide

WebSocket FT4/8

Overview

WebSocket FT4/8 is a browser-based FT8 and FT4 decoder. It connects to SDRconnect via WebSocket, receives demodulated audio, and decodes digital signals in real time — no installation required. Decoded messages are displayed with UTC time, SNR, frequency offset, and callsign details including country lookup. A live spectrum and waterfall display, station map, and band-hopping feature are also included.

NOTE	WebSocket FT4/8 runs on Windows, Mac, and Linux. It has been tested with Microsoft Edge, Google Chrome, and Firefox. Safari has not been tested.

Requirements

Before using WebSocket FT4/8, ensure the following are in place:

- SDRconnect running on your local network with WebSocket server enabled (see below).
- Microsoft Edge, Google Chrome, or Firefox browser.
- System clock synchronized to UTC within 1 second. Windows users should run a dedicated time sync tool such as NetTime, Meinberg NTP or JTSync. Mac and Linux have adequate time sync. Verify that timing is OK by going to https://time.is/ and seeing if clock is exact.

SDRconnect Setup

WebSocket FT4/8 receives audio from SDRconnect via its built-in WebSocket server. This must be enabled before connecting.

Enabling the WebSocket Server

- In SDRconnect, click the three-dot menu (···) in the top-right corner.
- Select Preferences from the menu.
- Toggle WebSocket Server on (see screenshot in Appendix).

NOTE	The WebSocket Server setting is remembered between SDRconnect sessions. You only need to enable it once.

Network Access

- Same computer: use the default IP 127.0.0.1 and port 5454.
- Local network: enter the IP address or name of the computer running SDRconnect. SDRconnect must be run with elevated privileges (Run as Administrator on Windows 11).
- Remote / internet: your router must forward port 5454 to the SDRconnect computer. Run SDRconnect as Administrator.

Getting Started

Follow these steps to begin decoding:

- Open WebsocketFT48_v1.html in Microsoft Edge, Google Chrome, or Firefox.
- Enter the IP address and port of the computer running SDRconnect (defaults: 127.0.0.1, port 5454). Previously used IP and port combinations are saved — click in the IP address field to see the history dropdown.
- Click Connect. The status indicator turns green when connected.
- Select a band from the Band drop-down menu.
- Click Decode Off to enable decoding — the button turns green and displays Decode On.
- Decoded messages appear in the decode window after each FT8 (15-second) or FT4 (7.5-second) cycle.

Help

Click the Help button at any time for a full description of all controls and features.

Toolbar Buttons

Button	What It Does
Decode Off / On	Starts and stops the FT8/FT4 decoder. When active the button turns green.
FT8 / FT4	Switches between FT8 and FT4 decoding modes.
Fast / Normal / Deep	Sets the decoder depth. Fast uses minimal CPU; Normal is the recommended default; Deep attempts more aggressive decoding at higher CPU cost.
Country Off / CQ / All	Controls whether country names appear alongside decoded callsigns. CQ shows countries on CQ messages only; All shows them on every decode.
Band	Selects the amateur radio band to tune to. Changing bands resets the decoder and retunes the SDR.
Hop Off / Hop On	Opens the Band Hopping configuration modal. When hopping is active, the Decode, Mode, and Band controls are locked.
Map	Opens the Station Map showing all decoded stations plotted on a world map for the current session.
AGC Off / On	Enables or disables Automatic Gain Control in SDRconnect. The recommended setting is AGC Off. SDRconnect AGC will be set back to On when disconnected.
Clear (Decode)	Clears the decoded messages list and resets the station map.
Clear (Spectrum)	Clears the spectrum and waterfall display.

Decode Window

Decoded messages appear after each cycle showing UTC time, SNR (dB), time offset (dt), audio frequency (Hz), and the decoded message text.

- Session totals for Decodes, Callsigns, and Countries are shown above the decode list.
- Left-click any message to open a split panel showing all messages decoded on that frequency.
- Right-click any message to open the sender’s QRZ page in a new browser tab.

Audio Level

Adjust the Audio Level slider so that the Audio Meter shows green bars reaching mid-scale. Avoid orange or red bars, which indicate clipping. The AGC Off button should normally remain in the off (default) state.

Spectrum & Waterfall

The spectrum display shows signal levels across the FT8/FT4 audio passband (0–3320 Hz). The waterfall below it shows signal history over time, with stronger signals appearing brighter.

Control	What It Does
Slow / Fast	Sets the waterfall scroll speed. Slow is easier to read during active decoding.
Cumulative	Toggles cumulative frame averaging for a smoother spectrum display.
Autoscale	Automatically adjusts the Min and Max dB levels to fit the current signal environment.
Min	Sets the minimum dB level of the colour scale. Signals below this level appear black.
Max	Sets the maximum dB level of the colour scale. Signals above this level appear at full brightness.
Clear	Clears the spectrum and waterfall display.

Station Map

Click the Map button to open the Station Map, which plots all decoded stations on a world map for the current session.

- Stations are plotted using their transmitted Maidenhead grid square when available, otherwise by country location.
- Map markers are colour-coded by band. A legend is shown above the map.
- Click any marker to see station details: callsign, locator or country, band, SNR, and time last heard in UTC.
- A QRZ button appears in the popup for resolved callsigns.
- Zoom with the mouse wheel; pan by clicking and dragging.
- The map resets when you clear the decode panel, disconnect, or change bands.

Band Hopping

Band Hopping automatically cycles through multiple bands, collecting decodes on each band for a configurable number of cycles before moving to the next.

- Click Hop Off to open the Band Hopping modal.
- Configure up to 11 bands with individual cycle counts and FT8/FT4 mode per band.
- Enable or disable individual bands using the checkbox in each row.
- Click Start Hopping to begin. The Decode, Mode, and Band controls lock while hopping is active.
- Each band tracks its own decode, callsign, and country counts. Click Reset Stats to clear them.
- Your band hopping configuration is saved automatically and restored the next time you open the app.

Appendix – Enabling the WebSocket Server in SDRconnect

To enable the WebSocket Server, click the three-dot menu (⋯) in the top-right corner of SDRconnect, open Preferences, and toggle WebSocket Server on.